Question Paper

Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV / DEC 2024

COMPUTATIONAL INTELLIGENCE AND ENVIRONMENTAL SUSTAINABILITY [ECE 4308]

Marks: 50 **Duration: 180 mins.**

DISCRIPTIVE

Answer all the questions.

Instructions to Candidates :
Answer ALL questions Missing data may be suitably assumed

1)		Discuss how computational intelligence (CI) techniques are being applied to promote environmental sustainability. Explain with example any one paradigm with its application in energy management	(5)
	A)		
	B)	Discuss the United Nations' Sustainable Development Goals (SDGs) and their significance in addressing global challenges. Give examples for initiatives being implemented to achieve these goals, and analyse the role of governments, public sector, and civil society in driving progress toward sustainability. What are the challenges to be faced in achieving these goals by 2030?	(3)
	C)	The McCulloch-Pitts neuron designed to implement the logical function F (A, B, C) =A BAR ·B + C Calculate sets of weights and and the threshold .	(2)
2)	A)	Consider a simple Artificial Neural Network (ANN) with two input neurons, one hidden layer with two neurons, and one output neuron. The network uses the sigmoid activation function. The input values are X1 =0.8 and X2 =0.5, and the target output is t=0.6. The initial weights of the network are: Weights from input to hidden layer W11 =0.1, W12 =−0.2, W21=0.4, W22=0.2 and Weights from hidden layer to output W01=−0.3, W02=0.5. The learning rate η is 0.1. Determine the following, show all intermediate calculations, and explain steps of backpropagation process. a) Perform a forward pass through the network and calculate the output of the ANN. b) Compute the error at the output layer using the mean squared error. c)Use the backpropagation algorithm to compute the gradients of the weights in the network and update the weight from hidden layer to output using gradient descent.	(5)
	B)	Given a 2-input neuron with logistic activation function with slope parameter of a=2. Let the input [-1 1], weight [0.1, 0.5] and output=0.73. compute the value of weight of bias input b1.	(3)
	C)	Realize two input NPT logic using McCulloch Pitts neuron model considering +1 as the bias value of the neuron.	(2)
3)	A)	Explain how Particle Swarm Optimization (PSO) algorithm can be applied to solve optimization problems related to the United Nations' Sustainable Development Goal , sustainable cities and communities (SDG 11), and discuss how PSO can contribute to achieving these goals. Provide examples of real-world applications and highlight the challenges and opportunities of using PSO in these contexts	(5)
	B)	Discuss QGIS and explain how an open-source Geographic Information System (GIS) tool, meets the needs of researchers, planners, and decision-makers in performing spatial analysis. Include an examination of specific use cases, such as disaster management, urban planning, and resource management, where spatial analysis has proven essential	(3)
	C)	Perform 4-point crossover for the following cross sites at 2, 5 ,9 ,10th place 0 0 1 0 1 1 0 0 1 1 1 0 10 1 and 1 1.0 0 1 0 1 1 0 1 0 1 1 0 1.	(2)
4)		Discuss the role of large language models (LLMs) in advancing AI tools and applications across	(5)

	A)	various industries. Explain the concept of prompt engineering and its significance in optimizing the performance of LLMs. Provide examples of how prompt engineering can enhance AI-driven tasks such as natural language processing, content generation, and decision-making. Additionally, analyze the challenges and ethical considerations with LLMs and prompt engineering in real-world scenarios	
	B)	Consider a university student who needs to create an optimized study timetable that balances their academic responsibilities with personal activities, such as gaming and sleep. Using a Genetic Algorithm (GA), outline the steps involved in developing an effective timetable that maximizes study efficiency without compromising the student's need for leisure and rest. Discuss the representation of potential solutions, the fitness evaluation criteria, the selection process, crossover and mutation methods, and how the GA can converge to an optimal solution. Provide examples of constraints that might be included and discuss the challenges faced during the optimization process	(3)
	C)	For the given two parent chromosomes P1= [1,0,1,1,0,1] and P2= [0,1,0,0,1,0], if a single-point crossover is applied after the third gene, determine the resulting offspring?	(2)
5)		Consider a function which takes a 2D input X = (X1, X2) and has 2 parameters w= (W1,W2) given by f (X,W) = σ (σ (W1X1)W2 + X2) where σ(X)= 1/ 1+e^-X . Use BP to estimate the right parameters value and start by setting both the parameters to 0. Assume that we are given a training point X1=1, X2 =0, y=5. Given this information answer the following: a) What is the value of f / w2 If the learning rate is 0.5, b) What will be the value of after one update using back propagation	(5)
	B)	Discuss how Multinational corporations (MNCs) such as Amazon utilize advanced optimization algorithms in machine learning and artificial intelligence to solve complex problems and enhance their operational efficiency. Analyse how it has implemented advanced optimization algorithms framework.	(3)
	C)	A 5-3-1 feed forward neural network architecture with one bias in input layer and one in hidden layer. Determine : a) Total Number of weights in the network. b) Also Sketch the network structure with 5 inputs [x1, x2, x3, x4, x5] = [0.8,0.6,0.4,0.2,0.1] and find out net-input to the output neuron, output Y for the network with binary sigmoidal activation function. Assume weights and biases.	(2)

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